

Solution Code - Module 3-Core Java - GradeCalculator

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q09_GradeCalculator.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q09_GradeCalculator {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.print("Enter marks obtained (out of 100): ");
7 |
8 |         if (scanner.hasNextDouble()) {
9 |             double marks = scanner.nextDouble();
10 |
11 |             if (marks < 0 || marks > 100) {
12 |                 System.out.println("Error: Marks must be between 0 and 100.");
13 |             } else {
14 |                 char grade;
15 |                 if (marks >= 90) {
16 |                     grade = 'A';
17 |                 } else if (marks >= 80) {
18 |                     grade = 'B';
19 |                 } else if (marks >= 70) {
20 |                     grade = 'C';
21 |                 } else if (marks >= 60) {
22 |                     grade = 'D';
23 |                 } else {
24 |                     grade = 'F';
25 |                 }
26 |                 System.out.println("Assigned Grade: " + grade);
27 |             }
28 |         } else {
29 |             System.out.println("Error: Invalid marks entered.");
30 |         }
31 |
32 |         scanner.close();
33 |     }
34 | }
```